# PROJECT BASED LEARNING (PBL) REPORT
# PHASE 2
## AI & ML Lab Using Python

# Credit Card Fraud Detection Using Machine Learning

**Problem Id: 12**


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## 1. Problem Description

Credit card fraud detection is a critical problem in the financial sector, where fraudulent transactions cause significant losses to banks and customers. With the increasing use of digital payments, detecting fraudulent activities has become more challenging. Fraudulent transactions are rare compared to legitimate ones, making the dataset highly imbalanced and difficult to model accurately.

The objective of this project is to develop a machine learning-based system that can classify transactions as legitimate or fraudulent. The system uses transaction-level features such as amount, time, location, and behavioral patterns to identify suspicious activities. The goal is to build a reliable model that can detect fraud effectively while minimizing false negatives.

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## 2. Architecture

The architecture of the system follows a structured machine learning pipeline:

Data Collection → Data Preprocessing → Train-Test Split → Feature Scaling → Imbalance Handling (SMOTE) → Model Training

**Steps Explanation:**

- Data Collection: A transaction dataset containing financial records is used.
- Data Preprocessing:
  - Removal of irrelevant columns (IDs)
  - Handling missing values
  - Encoding categorical features
  - Converting transaction time into useful format (hour)
- Train-Test Split: Dataset is divided into training and testing sets to evaluate model performance.
- Feature Scaling: Numerical features are scaled using StandardScaler for better model performance.
- Imbalance Handling: SMOTE is applied to the training data to balance fraud and legitimate transactions.
- Model Training: Machine learning models are trained on the processed dataset.

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## 3. Models Used

The following supervised machine learning models were implemented:

1. Logistic Regression

A linear classification algorithm used for binary classification. It predicts the probability of a transaction being fraudulent based on input features.

2. K-Nearest Neighbours (KNN)

A distance-based algorithm that classifies a transaction based on the majority class of its nearest neighbors.

3. Support Vector Machine (SVM)

A powerful classification model that finds the optimal boundary (hyperplane) between fraudulent and legitimate transactions.

These models were trained using the balanced training dataset and later evaluated for performance.

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## 4. Evaluation Metrics Used

The performance of the models was evaluated using the following metrics:

- Accuracy: Measures the overall correctness of the model.
- Precision: Indicates how many predicted fraud cases are actually fraud.
- Recall: Measures how many actual fraud cases are correctly identified.
- F1-Score: Harmonic mean of precision and recall.
- Confusion Matrix: Displays true positives, false positives, true negatives, and false negatives.

In fraud detection, recall is considered the most important metric because missing a fraudulent transaction can lead to significant financial loss.

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## 5. Preliminary Results and Visualizations

After training the models on the processed dataset:

- Logistic Regression and SVM showed strong performance with high accuracy.
- KNN showed comparatively lower accuracy due to sensitivity to data distribution and sample size.
- Initial results indicate that the models are capable of learning patterns from the dataset.
- Confusion matrix and evaluation metrics provide insights into model performance.

These preliminary results demonstrate that the system is capable of detecting fraudulent transactions, forming a strong foundation for further evaluation and optimization.

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## 6. References


1. Python Documentation – https://docs.python.org
2. Scikit-learn Documentation – https://scikit-learn.org
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4. Imbalanced-learn Documentation – https://imbalanced-learn.org
5. GeeksforGeeks – https://www.geeksforgeeks.org